

Confidence as Collateral

Strictly Proper Slashing for Accountable Automated Decisions

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This is a mechanism and a measured prototype, not a deployable protocol. *Only the calibration head is zero-knowledge proved — its input logit is unverified, dispute resolution rests on an admin-appointed N-of-M committee whose collusion carries the authority a single key would, and an unvalidated collusion detector can withhold a claimant's payout. Section 8 states each limitation in full; Section 9 states what would close each one, and records that one closure made the problem harder rather than easier.*

Abstract

Accountability mechanisms for automated decisions penalise error at a confidence-independent rate, so an operator is never punished for asserting 99% confidence in a decision it privately believes at 60%. We study slashing staked collateral by the **Brier score** over (reported confidence, adjudicated outcome). Because that score is strictly proper, expected loss is uniquely minimised when the operator reports its true subjective probability: honest confidence becomes the loss-minimising strategy rather than an assumption. We show the property survives fixed-point on-chain arithmetic, and prove that the slash cap must be 100% of stake — any lower cap makes maximal over-confidence optimal, so a prudential-looking parameter silently inverts the mechanism. Embedding the rule in a decision market yields a participation condition and a sharp negative: against a *single* buyer an optimally tuned flat bond achieves exactly the same welfare, and the separation appears only under buyer heterogeneity.

We report a measured prototype. Across 10 pinned seeds, temperature scaling cuts Expected Calibration Error from 0.1853 ± 0.0248 to 0.0870 ± 0.0218 on held-out UCI German Credit data, in every seed, replicating on a second 30,000-row dataset labelled with observed defaults; an EZKL/halo2 circuit proves the calibration head in 2.09 s and verifies on-chain for 684,696 gas, flat across a $16,897\times$ parameter increase. The central result is negative and is the paper's main claim: a proved calibration head is **not** sufficient for a trustworthy attestation, because the proof binds the map from logit to confidence and nothing else, so an operator that fabricates the logit obtains a proof that verifies. Three further measurements ran against the design — a statutory unbonding period at which capital lockup exceeds the slash eightfold, a subgroup gap that was estimator bias at one scale and real at another, and a detector reporting AUC 0.998 that flags 0.94% of honest claimants at the threshold its contract enforces. Every claimed weakness is executed as a passing test.

Keywords: proper scoring rules, mechanism design, zero-knowledge machine learning, calibration, staking and slashing, agentic payments.

1. Introduction

Automated credit decisions are entering a compliance regime that demands records rather than assurances. Under the EU AI Act, credit scoring is high-risk; Article 26 obliges deployers to monitor operation and retain logs for at least six months, and Article 86 establishes a right to explanation. These provisions mandate *logging and ex-*

planation. They do not establish that the logs are truthful, nor do they price the difference between a wrong decision made tentatively and a wrong decision made with asserted certainty. A deployer that logs a confidently wrong decision is compliant [1].

The gap. Existing accountability mechanisms penalise error at a rate independent of the confidence asserted. Chainlink staking slashes a fixed amount on an

SLA breach [3]; Nexus Mutual pays or refuses a claim by member vote [4]; HeLa's agent bonds slash "a fraction" on an adverse governance verdict [29]. Under any such rule, an operator asserting 99% confidence and one asserting 51% pay identically for the same wrong call — so nothing induces the 51%. Separately, a literature on proper scoring rules has known since Brier [9] and Gneiting and Raftery [10] exactly how to price a probabilistic claim so that honesty is optimal. The two have not been joined: scoring rules elicit beliefs from forecasters who are paid, not from operators who are *staked*.

This paper joins them, implements the result on-chain, and reports what happens when the implementation is measured rather than described. Six questions organise it. Slashing by Brier score *does* make truthful confidence loss-minimising, and the property survives fixed-point on-chain arithmetic (§3); the confidence-producing step *can* be proved in zero knowledge at 2.09 s and verified for 684,696 gas (§6.2); and proving cost is *flat* in the size of the proved head, so circuit capacity rather than parameter count binds (§6.2).

Two answers are negative, and they are the more useful results. A proved calibration head is **not** sufficient to make an attestation trustworthy (§5). And of five candidate enhancements ablated on an identical protocol, only one earns its place: a deeper base model is null and deep-ensemble epistemic uncertainty is actively harmful (§6.3). A sixth question — when attaching an attestation is worth its cost, and whether it dominates a simpler bond — is answered conditionally and partly against the mechanism (§4).

1.1 Contributions

1. A slashing rule on a strictly proper scoring rule, with the properness argument in closed form and its scope conditions stated as part of the claim.
2. **Proposition 2: the slash cap must be 100% of stake.** Below $x \geq \max(p, 1-p)$ the minimiser jumps to a boundary report, so a capped protocol *pays for* maximal overconfidence — limited liability's risk-shifting effect arriving through a parameter that looks prudential. This also settles Brier over the log score, which is unbounded and cannot be paid from finite stake.
3. A welfare model of a decision market (§4) yielding a participation condition and an equivalence result that bounds the mechanism's own advantage.
4. A working implementation: fixed-point Brier arithmetic at 543 gas, an EZKL/halo2 circuit, and an attestation contract that rejects unverified proofs — with a two-dataset evaluation over 10 pinned seeds.
5. **Four negative results**, reported at the weight of the positive ones: the insufficiency of a proved head (§5), an unaffordable statutory unbonding period (§7.1), a subgroup gap that was estimator bias at one scale and real at another (§7.2), and a detector whose headline AUC does not describe its deployed operating point (§7.3).

2. Related work and positioning

zkML. EZKL [6] compiles ONNX graphs to halo2 circuits, and the field's practical constraint — proving cost dominated by circuit capacity rather than model size — is the design constraint this work is built around: prove the small calibration head, not the classifier. The consequence, that everything outside the proved subgraph is unbound, is the subject of §5.

Staking, and scoring rules. Chainlink [3], Nexus Mutual [4] and Kleros [7] stake collateral against adjudicated outcomes, none weighting the penalty by stated confidence. Modern networks are overconfident and temperature scaling largely fixes it [11]; the Brier score [9] is strictly proper [10]. Foresight Arena [5] scores on-chain forecasts by Brier but does not stake them.

Agentic payments. A body of work postdating this project targets the same problem from the payments side, and its distribution is itself a finding: it is almost entirely about whether payment is correctly bound to *delivery* [25, 26, 27], and almost not at all about whether the delivered decision was any good. ERC-8004 [23] standardises an agent reputation registry; the first empirical study of it [24] found that across three chains only 3–15% of registrations expose a live endpoint while 59–91% of reviewers show coordinated Sybil behaviour, concluding the registry "cannot function as a trust signal." That is the strongest external evidence for what this mechanism does, and it must be cited as evidence rather than paraphrased as endorsement: Brier *relocates* the grounding assumption from "anonymous reviewers are honest" to "the resolver committee is not colluding," which is smaller and more auditable but is not the elimination of trust.

Table 1: Positioning. *Conf* = the penalty is weighted by the reporter's own stated confidence; *Stake* = collateral is at risk; *ZK* = the scored quantity is zero-knowledge proved; *x402* = deployed against the agentic-payments rail today.

Work	Conf	Stake	ZK	Adjud.	x402
Chainlink staking [3]	–	yes	–	alerting	–
Nexus Mutual [4]	–	yes	–	member vote	–
Kleros [7]	–	yes	–	juror pool	–
Foresight Arena [5]	yes	–	–	market	–
EZKL [6]	–	–	yes	–	–
ERC-8004 [23]	–	–	–	client feedback	yes
TraceRank [28]	–	–	–	payment graph	yes
HeLa bonds [29]	–	yes	–	DAO vote	–
This work	yes	yes	yes	N-of-M	–

The final row is the only one carrying both *Conf* and *Stake*, and that conjunction is the novelty claim. It is also

not x402-native, and the column exists so that shows rather than hides: three of the rows above are deployed against real traffic and this is a prototype.

3. The mechanism

Definition 1 (Confidence-calibrated slash). Let S be the collateral an operator has staked, $c \in [0,1]$ the confidence it reported that its decision was correct, and $o \in \{0,1\}$ the adjudicated outcome of a dispute over that decision. On an upheld dispute the contract slashes $\text{slash} = S \cdot (c - o)^2$, subject to a protocol maximum fraction of S .

The penalty is the Brier score of a single forecast, denominated in stake. Everything that follows depends on one property of that score and nothing else about it. Write p for the operator's private belief that its decision is correct — unobservable, and the quantity the mechanism must elicit.

3.1 Truthful reporting minimises expected loss

Assumption 1 (Truthful adjudication). The adjudicated outcome o equals the true correctness of the decision, and whether a decision is disputed is independent of the confidence reported.

Proposition 1 (Strict properness of the Brier slash). *Under Assumption 1, an operator reporting c faces expected slash $E[\text{slash}] = S ((c - p)^2 + p(1 - p))$. This is uniquely minimised at $c = p$, and every misreport costs $S(c - p)^2$ in expectation.*

Proof. The dispute is upheld with probability p , giving slash $S(c - 1)^2$; otherwise the slash is Sc^2 . Hence $E[\text{slash}] = S (p(c - 1)^2 + (1 - p)c^2) = S (c^2 - 2pc + p)$. Completing the square gives $S ((c - p)^2 + p(1 - p))$. The bracket is strictly convex in c with its only stationary point at $c = p$, and the residual $p(1 - p)$ does not depend on c . $\square$

Two consequences. Overclaiming is not merely unrewarded but priced quadratically, so shading slightly is cheap while asserting near-certainty about a coin flip costs nearly the whole stake. And the residual $p(1 - p)$ is irreducible: a well-calibrated operator that is sometimes wrong still pays, which is what makes the rule a price rather than a punishment. The operator need not be honest, only selfish.

3.2 The cap must be 100%, and why the Brier score

A protocol that slashes stake will be tempted to cap the penalty. It must not.

Proposition 2 (Capping the slash destroys strict properness). *Let the slash be $\min(S(c - o)^2, \kappa S)$ for a cap $\kappa \in (0,1]$. Truthful reporting remains optimal iff $\kappa \geq \max(p, 1 - p)$. For κ below that bound the expected-loss minimiser is a boundary report: $c = 0$ when $p < \text{frac}12$, and $c = 1$ when $p > \text{frac}12$.*

Proof. The cap binds exactly where $(c - o)^2 > \kappa$. For $p > \text{frac}12$ and $\kappa < p$, reporting $c = 1$ caps the loss on the $o = 0$ branch at κS while paying nothing on the $o = 1$ branch, giving expected loss $(1 - p)\kappa S$. The truthful report pays $S p(1 - p)$, and $(1 - p)\kappa S < S p(1 - p)$ whenever $\kappa < p$. The case $p < \text{frac}12$ is symmetric. $\square$

This is limited liability's risk-shifting effect arriving through a parameter that looks prudential: a capped protocol *pays for* maximal overconfidence. Verified numerically and pinned by three tests against the deployed `BrierMath.sol`, whose `maxSlashBps` is 10,000.

It also settles the choice of scoring rule. The logarithmic score is strictly proper too, but unbounded — an operator confidently wrong owes infinity, which no finite stake can pay, and truncating it triggers exactly Proposition 2. The Brier score is the natural strictly proper rule bounded on $[0,1]$.

Scope conditions, stated as part of the claim rather than as caveats: the circuit takes the base model's logit as an *unverified* public input, so a fabricated logit yields a proof that verifies (§5); nothing forces the operator to feed the circuit the logit its deployed model produced; and if only rejections are ever disputed, Assumption 1 fails and the loss surface is asymmetric in a way not analysed here.

4. An economic framework for decision markets

§3 shows the slash elicits truthful confidence. It does not show anyone should want it. A buyer-agent purchases a decision through a priced endpoint and must act on it: acting on a correct decision yields V , acting on a wrong one costs K , abstaining yields 0. A rational buyer acts iff its posterior exceeds $t = K/(V + K)$ — the buyer's threshold, determined entirely by its own payoff asymmetry and **private to it**.

Proposition 3 (Uninformative reporting without a penalty). *With no penalty attached to the report, every c yields the seller the same payoff and $c = 1$ weakly dominates. The buyer's posterior is therefore independent of c .*

Costless signals are uninformative signals — the same gap the ERC-8004 study [24] measured empirically. Attaching Definition 1 makes c a sufficient statistic for p , and the buyer can act iff $c \geq t$.

Proposition 4 (Participation condition). *Let $G = E_F[\max(0, pV - (1 - p)K)] - E_F[pV - (1 - p)K]$ be the screening value. Attaching an attestation at per-decision cost g raises welfare iff $g < G$, and the gain is exactly $G - g$.*

G is the value of being able to decline, and it is bounded — a mechanism costing more than the mistakes it prevents is not worth running. With a Beta(5,2) decision population (mean 0.714, matching the prototype's measured accuracy) and $V = 100$, $K = 400$, $G = 56.19$.

Table 2: The participation condition at measured per-attestation costs. Welfare is per decision, in the buyer's units.

Setting	g	Welfare	Gain	Verdict
L1, busy (30 gwei)	\$79.86	-66.58	-23.67	does not pay
L1, quiet (10 gwei)	\$26.62	-13.34	+29.57	pays
L2, typical (0.05 gwei)	\$0.13	+13.15	+56.06	pays

On a busy L1 the mechanism is welfare-negative.

The relevant comparison is not gas against the decision's price but gas against G . The condition also says when the mechanism is pointless for reasons unrelated to cost: $G \rightarrow 0$ when payoffs are near-symmetric and most decisions are worth acting on. Confidence attestation is worth paying for exactly where decisions are consequential, asymmetric, and genuinely uncertain — a narrower market than "all agentic commerce."

4.1 Against a flat bond, in the same model

HeLa's accountability bond [29] slashes a fraction q of a bond B at dispute rate d . The expected penalty $q B d(1-p)$ **does not depend on c** , so the reporting margin is untouched and Proposition 3 still applies. What the bond constrains is the participation margin: the seller sells only when $q > q B d(1-p)$, i.e. when $p \geq 1 - q/(q B d)$.

Proposition 5 (Equivalence against a single buyer). *For any single buyer with threshold t , the bond level $q B d = q/(1-t)$ achieves exactly the same welfare as the Brier mechanism, gross of attestation cost.*

Proof. The bond induces seller-side threshold $p_{\min} = 1 - q/(q B d)$; setting $q B d = q/(1-t)$ gives $p_{\min} = t$, so both mechanisms act on exactly $\{p \geq t\}$. ■

The numbers confirm it: both attain 13.2757. **Identical, not merely close.** This cuts against the mechanism and is stated plainly — for a single buyer, confidence elicitation buys nothing a correctly tuned bond does not. What it buys is robustness to mis-tuning (at half the optimal bond, welfare falls to -6.23), which is a weaker claim than the positioning table alone would suggest.

Proposition 6 (Pooling loss of threshold mechanisms). *With buyers heterogeneous in t , the Brier mechanism attains $E_j E_F[\max(0, p V_j - (1-p) K_j)]$, while any flat bond attains at most $\max_q E_j E_F[\mathbb{I}\{p \geq q\}(p V_j - (1-p) K_j)]$. The first weakly dominates, strictly whenever two buyers have distinct thresholds.*

Proof. The report $c = p$ is a sufficient statistic, so each buyer j attains its first-best set $\{p \geq t_j\}$. A bond fixes one pooled threshold q for all buyers; where $t_j \neq q$, buyer j either acts on decisions with $p < t_j$ or abstains on decisions with $p \geq t_j$, and either strictly lowers its payoff. □

With four buyers whose thresholds span 0.50–0.95, an optimally tuned bond attains 4.67 against Brier's 15.83 — an efficiency loss of **70.5%**. The loss is not mis-tuning; the bond is at its optimum. It is the structural cost of compressing a continuous signal into one binary sell/don't-sell rule.

This is the mechanism's actual economic claim, and it is narrower than "accountability." A bond communicates *the seller declined to sell you bad decisions at the protocol's threshold*; an attestation communicates *here is how good this decision is, apply your own threshold*. The second is more valuable exactly when buyers differ, and worth nothing when they do not. Buyer heterogeneity is modelled here, not measured.

5. System, and the boundary of the proof

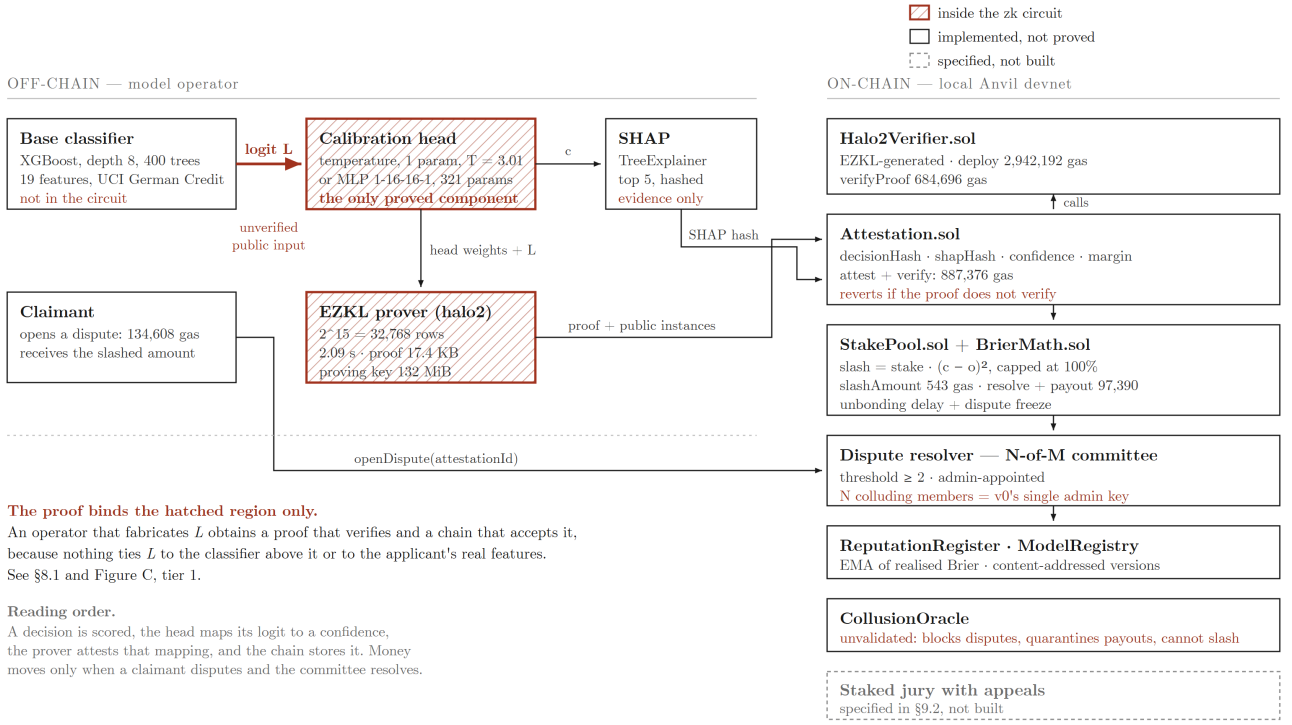


Figure 1: Component architecture, with measured cost at each step. Hatching marks the only subgraph inside the zk circuit; dashed outlines are specified but unbuilt. The heavy arrow is the base model's logit, which the circuit takes as an unverified public input — the boundary this section turns on.

An XGBoost classifier produces a logit L ; a one-parameter temperature head maps it to a confidence c ; EZKL proves *that mapping only*; **Attestation.sol** stores the record and reverts if the proof does not verify; **StakePool.sol** computes the slash on resolution. The base classifier and the SHAP explainer sit outside the circuit by design — the vectors they produce are hash-committed as evidence, which establishes that the operator recorded a particular explanation and nothing about whether it is faithful.

Theorem 1 (Scope of the tier-1 guarantee). *The zk proof establishes that a committed head with verifying key vk maps the public input L to the attested c . It establishes nothing about the provenance of L .*

An operator that fabricates L obtains a proof that verifies and a chain that accepts it. This is not a defect to be patched but a statement of what the cryptography buys, and it is asserted by a passing test (`test_tier1_marginIsUnverifiedOperatorSuppliedInput`) rather than conceded in prose. Three trust tiers follow:

- **Tier 1 — cryptographic.** Calibration-head execution and slash arithmetic. Holds against a fully malicious operator, and covers nothing else.
- **Tier 2 — economic.** Honest reporting (Proposition 1) and stake availability. Enforced in code by a two-step withdrawal behind an unbonding delay, with any open dispute freezing execution — but released by a tier-3 action.
- **Tier 3 — bounded trust.** Who loses money. An admin-appointed N-of-M committee whose N collud-

ing members carry exactly the authority v0's single admin key had. Reputation, being computed from tier-3 outcomes, inherits this exactly.

The cheapest attack today requires breaking no cryptography: attest over a fabricated L , or corrupt N committee members.

6. Results

All figures are produced by scripts in the repository. Environment: Windows 11, Python 3.13.5, xgboost 3.1.2, torch 2.9.1, EZKL 23.0.5, Foundry 1.7.1, laptop CPU, no GPU.

6.1 Calibration, on two datasets

Seeds are pinned in `config.EVAL_SEEDS` and the full list is always run, so a favourable subset cannot be reported selectively.

Table 3: Calibration across 10 pinned seeds, UCI German Credit. The train-fitted control is worse than not calibrating at all, in every seed.

Head	ECE	Brier	Accuracy
Uncalibrated	0.1853 $\pm$ 0.0248	0.2060 $\pm$ 0.0191	0.7505 $\pm$ 0.0209
Temperature (1 param)	0.0870 $\pm$ 0.0218	0.1758 $\pm$ 0.0113	0.7505 $\pm$ 0.0209
MLP (321 params)	0.1055 $\pm$ 0.0234	0.1908 $\pm$ 0.0119	0.7330 $\pm$ 0.0199
<i>Control: fitted on TRAIN</i>	<i>0.2434 $\pm$ 0.0212</i>	—	—

Learned $T = 3.47 \pm 0.45$, with $T > 1$ in every seed — the base model required softening everywhere. Accuracy is unchanged, as it must be: a monotone rescaling cannot move the decision boundary. Calibration buys a better-priced confidence, not a better classifier.

The 321-parameter MLP head does not beat the 1-parameter head, and the strength is reported as the data supports it: temperature wins in **7 of 10** seeds, Wilcoxon $p = 0.037$ — a *majority* result, not a uniform one, where a single run per head would have supported the stronger and less accurate claim.

The result replicates. Running the identical protocol — same splits, seeds, head, fitting, metrics, and deliberately the same base-learner hyperparameters — on UCI *Default of Credit Card Clients* (Taiwan, 30,000 rows, labelled with **observed defaults** rather than an analyst's credit grade):

Claim	German Credit	Taiwan default
ECE reduced in every seed	10/10	10/10
Mean ECE reduction	52.8%	54.8%
Learned $T > 1$	3.47 ± 0.45	2.80 ± 0.06
Realised Brier — what is slashed	0.2060 $\rightarrow$ 0.1758	0.1642 $\rightarrow$ 0.1490

The label change carries the weight: a finding that held on adjudicated opinion but not on realised outcomes would be much weaker. Hyperparameters were not retuned, so the base model is under-fit on the larger corpus and absolute accuracy is not the comparison to read — retuning would have tested whether the data can be modelled well, not whether the finding travels.

6.2 Proving and on-chain cost

Proving cost is **flat across a 16,897 $\times$ parameter increase**: mean 2.13 s, sd 0.09, regression slope +0.009 s per decade, Spearman -0.188 at $p = 0.603$. Rows used scale linearly with parameters (0.98 rows/param, $r = 0.992$), so fixed lookup-argument and column overhead dominate the multiply-accumulate count entirely.

The boundary — where cost steps past logrows 15 — is unmeasured, and we say so rather than extrapolating.

Table 4: Measured gas. The Brier arithmetic that is the substance of the mechanism is 0.08% of the per-decision cost.

Operation	Gas
Halo2Verifier deployment (one-off)	2,942,192
verifyProof (real EZKL proof)	684,696
attest (verify + store)	887,376
openDispute	134,608
resolveDispute incl. slash + payout	97,390
BrierMath.slashAmount (pure)	543

End-to-end on a local chain, confident-and-wrong costs 3,667 $\times$ confident-and-correct and 3.49 $\times$ uncertain-and-wrong — the ordering the mechanism exists to produce.

6.3 Five enhancements, ablated

Each candidate ran under the identical seed protocol and was kept only if it earned its place.

Enhancement	Effect on the priced quantity	Verdict
Conformal prediction	Distribution-free coverage, no measurable proving cost	kept
Deeper base model	Null	rejected
Deep-ensemble epistemic uncertainty	Degrades calibration	rejected
Counterfactual evidence	Evidence only, not proved	kept, tier 1
Collusion detection	Synthetic rings only	kept, unvalidated

The deep-ensemble result is the clearest: epistemic uncertainty is standard practice and it made calibration *worse*, interpretable only because a capacity control was run alongside — the architecture helps, the epistemic signal destroys the gain.

7. Three measurements that ran against the design

Reported at the same weight as the results that favour the mechanism, because a document reporting only its confirmations is not evidence of much.

7.1 The unbonding period is unaffordable

The period must exceed the time a claimant needs to notice a bad decision and act. For US consumer credit that is statutory, not estimable: 60 days for the consumer's free-file window (15 U.S.C. § 1681j, cited in § 1681m(a)), 30 days for reinvestigation (§ 1681i(a)(1)(A)), and 15

days for the permitted extension (§ 1681i(a)(1)(B)) — $\tau = 105$ days, fifteen times the value used in tests. Each term is a statutory maximum, the right posture for a security parameter.

Re-running the carrying cost $r \tau S$ at $r = 5\%$ inverts the earlier conclusion. At 7 days the carry is 0.0959% per cycle; at 105 days it is **1.4384%**, exceeding the expected slash (realised Brier 0.1758 times the dispute rate) for **any dispute rate below 8.18%** — roughly $8\times$ at 1%.

This weakens the paper's own incentive argument: if the slash is an eighth of the total cost, the 15% saving from calibration is about 1.7% of what participation costs. τ is pinned by statute above and by capital cost below, and the bounds do not meet, so closing the gap needs a different instrument — bonded insurance, rolling tranche release, or an at-risk fraction below the full stake — none of which is a parameter change.

7.2 A subgroup gap that was noise, then real

Aggregate ECE averages over the population, so it is blind to a defect confined to a subgroup — the sharpest scientific gap in the mechanism. The plan was to construct the adversary, show aggregate ECE misses it, and track a within-group variant on-chain.

The first-order result looked like confirmation: within-group ECE exceeds aggregate in **10 of 10** seeds, 0.1349 against 0.0870. **It does not survive a control.** Permuting subgroup labels while holding group sizes fixed reproduces the entire effect — null gap 0.0559 against the real partition's 0.0548, Wilcoxon $p = 0.92$, the real partition beating its own null in 3 of 10 seeds.

The cause is a small-sample bias in ECE itself. Within-bin accuracy is estimated from a handful of points and $|acc - conf|$ is an absolute value, so sampling error accumulates instead of cancelling. A model calibrated *by construction* ($p \sim U(0,1)$, $y \sim \text{Bernoulli}(p)$) scores:

n	68	106	200	500	2,000	10,000
ECE	0.1188	0.0989	0.0695	0.0451	0.0223	0.0098

The test split is 200 rows and the measurable subgroups hold 68 and 106. **At $n = 68$ the estimator's own bias (0.1188) exceeds the pipeline's aggregate ECE (0.0870).**

On the 30,000-row dataset, where subgroups hold 2,336–3,664, the same analysis resolves it: the effect **survives its permutation null** (gap 0.0067 vs 0.0042, 8/10 seeds, $p = 0.037$) and is about 12% of aggregate ECE. Both halves matter. The effect exists, *and* the first measurement of it was noise — reporting only the first would have overstated the problem sevenfold (an apparent 0.0479 against a measured 0.0067); reporting only the second would have missed that the instrument was what was being measured.

`SubgroupReputationRegister.sol` was **not built** either way, because binned ECE's bias at realistic group sizes is twenty times the effect it would enforce on. Tests

pin the null against erosion: shrinking the minimum group size, dropping the permutation control, or letting p fall below 0.05 all fail loudly.

7.3 A detector whose AUC does not describe it

The collusion detector reports **AUC 0.998** on injected rings and is wired to withhold a claimant's payout. Its false-positive rate on real traffic is unobtainable — the protocol has never run. But on synthetic traffic containing *no rings*, every flag is false by construction, and at the threshold the contract actually enforces (`MIN_SCORE = 0.8e18`, below which `flag()` reverts):

Traffic	FPR	Recall	FDR
No rings present	0.94% (95% Wilson upper bound 1.51%)	—	—
Rings at intensity 0.85	2.41%	87.2%	19.9%
Rings at intensity 0.40	1.79%	16.1%	50.0%

17 of 1,800 honest claimants flagged in a world with no collusion in it. One flagged claimant in five is honest at the easiest setting; at the hardest the detector recalls 16% while half of everyone it flags is innocent.

AUC is threshold-free and integrates over operating points the contract cannot reach. The deployed system has one fixed operating point, and its precision there is what governs whether attaching the oracle is defensible. Reporting a threshold-free metric while enforcing at a fixed threshold is how a system comes to look validated for something it is not. The recommendation is explicit: `pass address(0)`.

This does not close the limitation: synthetic traffic from a generator whose realism is itself an assumption clears the lowest available bar. What changes is that an operator now has a measured number where there was none.

8. Limitations

Stated as the security model rather than as omissions from it. Five capabilities an adversary retains today, each demonstrated by a passing test:

Adversary capability	Covered	Mechanism, or reason not
Forge a proof for a head it did not run	yes	halo2 soundness; 4/4 tamper checks
Alter an attested confidence after the fact	yes	stored on chain, proof-bound
Substitute weights under a claimed version	yes	content hash
Exit stake ahead of a pending dispute	yes	unbonding delay + dispute freeze
Act alone as a single resolver	yes	N-of-M threshold
Fabricate the input logit L	no	tier 1 binds the head, not the pipeline
Corrupt N of M resolvers	no	equals v0's single-key power
Exit before any dispute is raised	no	a bound on the dispute window, not the lock
Misreport within a protected subgroup	no	measurable only at scale (§7.2)
Flag an honest claimant	no	detector unvalidated on real traffic (§7.3)
Train dishonestly, register truthfully	no	content integrity $\neq$ training integrity

Beyond these: **ground truth for a rejection is unobservable** — a rejected applicant never demonstrates repayment, so "the decision was wrong" has no on-chain referent, and moving from one voter to N does not create a fact that did not exist. This is unsolved, not merely unimplemented.

Empirical scope. Two datasets, but both credit, tabular, binary, and run on one model family — nothing here separates "this calibration result holds generally" from "this holds for gradient-boosted trees." Calibration drift over time is unevaluated. Staking parameters are demonstration values with no actuarial basis, and correlated tail risk is not modelled.

Vocabulary. docs/PHASE3B_TRUST_MODEL.md fixes the vocabulary for describing dispute resolution, and was written before the code it describes. tests/test_claim_vocabulary.py greps this document for language attributing properties the system lacks — the forbidden list lives in the test rather than in prose, so a later edit cannot quietly upgrade a claim.

9. Open problems

Ordered by what blocks what. Two items are recorded as closed by measurements in this paper, and every item that was actually executed came back different from how it was specified — which is the pattern worth noting.

1. **Input-logit provenance.** Three routes, in decreasing order of guarantee: prove the base classifier in-circuit (prohibitive for depth-8, 400 trees); commit to the feature vector and weights, making fabrication *detectable* rather than impossible (cheap, closest to shipable); or attest the base model in an enclave (weakest — hardware trust replacing cryptographic trust).
2. **A defensible dispute layer.** A staked jury with an evidentiary standard and appeals. Blocked on a definition of the adjudicated event that does not yet exist.
3. **Decoupling the dispute window from capital lockup** — opened by §7.1, since no value of τ is satisfactory.
4. **A debiased calibration estimator**, characterised at the group sizes a deployment will see — opened by §7.2, and blocking any subgroup slashing rule.
5. **A second model family, and drift.** Now the more glaring empirical gap, and cheaper than the replication already done.
6. **Real labelled disputes**, without which §7.3's limitation cannot close.

Explicitly out of scope. Non-deterministic decision classes such as LLM outputs are *substantially harder, not a natural extension*: free-form generation has no canonical scalar confidence, no binary ground truth, and no stable notion of "the same decision" across sampling seeds. Applying a proper scoring rule there requires first defining the event space being scored, which is a research problem rather than an engineering task.

10. Conclusion

The mechanism-design claim is supported: slashing by a strictly proper scoring rule makes honest confidence loss-minimising, the property survives fixed-point on-chain arithmetic, and it is cheap — 543 gas for the arithmetic itself, 2.09 s to prove the head. Proposition 2 adds a condition a deployer would otherwise get wrong, since capping the slash looks prudential and inverts the mechanism.

The systems claim is **not** supported. A proved calibration head is not sufficient for a trustworthy attestation: the proof binds the map from logit to confidence and nothing else, so an operator that fabricates the logit obtains a proof that verifies. Dispute resolution remains an admin-appointed committee whose collusion carries the authority a single key did.

That invariance is the most durable result here. A better base model, a stronger uncertainty quantifier, a richer evidence bundle, a reputation history and a version registry all now sit on top of the same unproven input. **Sophistication above the trust boundary does not move the trust boundary**, and this work is a fairly thorough demonstration of that.

The economic answer is narrower still. Attaching an attestation pays only where verification costs less than

the errors it screens out — at measured L1 prices it does not — and against a single buyer an optimally tuned flat bond achieves *exactly* the same welfare. The separation is real but specific: a bond compresses quality into one pooled threshold, while an attested confidence is a sufficient statistic each buyer applies its own threshold to, worth 70.5% of the available gain across a heterogeneous buyer mix. That heterogeneity is modelled and not measured.

The three measurements of §7 ran against the design and are reported at full weight. Two of them are cautions about instruments rather than about the mechanism — an estimator whose small-sample bias exceeded the effect it was measuring, and a threshold-free metric that did not describe the fixed threshold its contract enforces. Both are mistakes a careful reader of the earlier literature could have made.

The most useful result may be the negative one: it establishes precisely which parts of "verifiable AI accountability" the cryptography actually buys, and which parts remain a matter of whom you trust.

Reproducibility

Every number is produced by a script; none is transcribed by hand. This sequence was executed against a fresh `git clone`, which found a real defect — `contracts/lib/` is not committed, so `forge test` failed until the `forge install` line below was added. With it, the clean checkout runs **144 Solidity and 113 Python tests green** and regenerates the market-model and subgroup artifacts **byte-identical**.

```

pip install -r requirements.txt
bash scripts/00_setup_tools.sh
cd contracts
forge install foundry-rs/foundry-std@v1.16.2 --no-git
cd ..
python scripts/10_train_calibrate.py
python scripts/12_multiseed_eval.py
python scripts/31_zk_prove.py
python scripts/23_second_dataset.py
python scripts/22_market_model.py
python scripts/21_subgroup_adversary.py
python scripts/24_detector_fpr.py
cd contracts && forge test
python -m pytest tests/ -q

```

In order: EZKL v23.0.5 and the un-committed Foundry library; calibration with its ECE gate and the 10-seed pass (§6.1); circuits and proofs (§6.2); the second-dataset replication (§6.1); the market model, which produces every number in §4; the subgroup null (§7.2); the detector's false-positive rate at the enforced threshold (§7.3); then both suites.

Two limits on that verification, stated so it is not read as more than it is: it ran on the development machine and OS, so it establishes self-containment rather than

portability; and the zkML stages were not re-run in the clean checkout, so §6.2's proving figures come from committed artifacts rather than a fresh proving run.

`ThreatModel.t.sol` does **not** assert the system is secure — it asserts that each documented weakness is real and reachable, so a failure there means the threat model has drifted from the code and this paper is wrong.

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Statutory sources for §7.1. 15 U.S.C. § 1681i(a)(1)(A) and (B); 15 U.S.C. § 1681m(a). <https://www.law.cornell.edu/uscode/text/15/1681i>